# MICHAEL FRYER



## Data Scientist & Software Engineer | M.Sc. Applied Data Science Candidate


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Aufenthaltstitel Holder | English (Native) | German (A2)
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## SUMMARY


*Software engineer transitioning into data science, currently pursuing an M.Sc. in Applied Data Science at Frankfurt School of Finance & Management. Experienced in Python-based machine learning, statistical analysis, and data visualization, with a professional background in full-stack software development. Seeking to apply analytical and engineering skills to data-driven roles in industry.*


## EDUCATION

### M.Sc. in Applied Data Science
September 2025—Present
**Frankfurt School of Finance & Management**
Frankfurt am Main, Germany
Expected graduation: Summer 2027 | Relevant coursework: Machine Learning 1 & 2, Deep Learning, NLP, Computational Statistics & Probability, Databases & Cloud Computing

### B.Sc. in Software Engineering and B.Sc. in Computer Science
August 2015—May 2019
**Montana Tech**
Butte, MT, USA

## EXPERIENCE

### Quantitative Analyst - Intern
August 2026—Present
**Deutsche Börse Group**
Eschborn, Germany
- Developing novel business solutions for the Market Data and Services division, focusing on speed and reliability of data delivery to clients.

### Associate Game Programmer
October 2019—July 2021
**Heavy Iron Studios**
Manhattan Beach, CA, USA
- Worked closely with artists to transform their UI mock-ups into a functional user experience.
- Extended Unreal Engine's replication graph to meet our network performance requirements.
- Shipped production-ready features across multiple Unreal Engine 4 patch releases in preparation for a full commercial game launch.
- Built and deployed the Live Operations portal using Django and AWS, enabling real-time content management for a live-service title post-launch.

### Software Engineer
September 2018—August 2019
**Butte Food Bank**
Butte, MT, USA
- Co-developed FRED in a three-person team, a full-stack inventory and reporting system that streamlined operations for the Butte Food Bank.
- Developed the front-end application of FRED using WPF Forms.
- As sole contractor after graduation, expanded reporting features to help the Food Bank meet government regulations.

## PROJECTS

### Urban Mobility Risk Analysis *| Python, Pandas, DBSCAN, Streamlit*
Applied DBSCAN clustering to 2.4M geolocated injury accidents across Germany (2016–2025) to identify and rank high-risk zones in any of its 401 districts. Built a public Streamlit dashboard with a district picker, outlier filtering and per-district parameter tuning by binary search.

### Agentic Connections *| Python, Anthropic API, pytest*
Red-teaming A/B experiment on a tool-use LLM agent that plays NYT Connections and reports its own results, checked against the game engine's ground truth. 300 pre-registered runs across two Claude models, analyzed with Fisher's exact tests; offline test suite against a fake API client.

## TECHNICAL SKILLS

Machine Learning | Neural Networks | Feature Engineering | Data Analysis | Statistical Analysis | Data Visualization | EDA | Python | PyTorch | Scikit-learn | Pandas | NumPy | SciPy | SQL | Plotly | Jupyter | Claude Code | AWS | Docker | Linux/Bash | Git | TypeScript | C/C++

Analytical Thinking | Problem Solving | Attention to Detail | Communication | Collaboration